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## Decision Making Study

1 message

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Sun, May 24, 2026 at 8:11 PM

<http://localhost:4279/>

# Decision Making Study

PSY 120L Reproducible Analysis Report

## About This Report

*This report will analyze the following data collected from our ecision Making study.*

## Study Overview

*This study will use a  $2 \times 2$  between-subjects experimental design. The primary independent variable is perceived social norms regarding risky behavior, with two experimental conditions: ENGAGE and AVOID. In the ENGAGE condition, participants will be told that college students commonly engage in a variety of risky behaviors. In the AVOID condition, participants will be told that college students generally avoid risky behaviors. Participants will be randomly assigned to one of the two conditions. The moderator variable is level of conscientiousness (high vs. low), measured using items adapted from the Big Five Inventory. Conscientiousness is*

## Setup and Software Environment

*add snippet here*

### Load Packages

Show code

```
library(tidyverse) # data manipulation + plotting
library(psych)     # scale construction + reliability
library(afex)      # ANOVA models + effect sizes
library(emmeans)   # estimated marginal means + confidence intervals
```

## Data Source and Analysis Dataset

*The data was derived from undergraduate students from th University of California's PBS department. The final data set, after data cleaning, was sored as .csv file and importend into R for reproducible data.*

## Import Dataset

## Show code

```
study_raw <- read_csv("data/decision_making_data.csv")
```

## Check Imported Dataset

Show code Show code

```
[1] "StartDate"      "EndDate"      "Status"
[4] "IPAddress"      "Progress"      "Duration (in seconds)"
[7] "Finished"       "RecordedDate" "ResponseId"
[10] "ExternalReference" "LocationLatitude" "LocationLongitude"
[13] "DistributionChannel" "UserLanguage" "consent"
[16] "mc"             "RTL_1"        "RTL_2"
[19] "RTL_3"          "RTL_4"        "RTL_5"
[22] "PNR_1"          "PNR_2"        "PNR_3"
[25] "PNR_4"          "PNR_5"        "con_1"
[28] "con_2"          "con_3"        "con_4"
[31] "con_5"          "con_6"        "con_7"
[34] "con_8"          "con_9"        "con_10"
[37] "con_11"         "con_12"       "age"
[40] "academic_year"  "gender"       "condition"
```

Show code

```
# A tibble: 6 × 42
  StartDate EndDate Status IPAddress Progress Duration (in seconds...1 Finished
  <chr>      <chr>   <dbl> <chr>      <dbl>          <dbl>   <dbl>
1 5/5/26 20:03 5/5/26... 0 72.205.8... 100          146      1
2 5/6/26 10:41 5/6/26... 0 169.231.... 100          298      1
3 5/6/26 14:03 5/6/26... 0 169.231.... 100          874      1
4 5/6/26 15:03 5/6/26... 0 169.231.... 100          211      1
5 5/5/26 19:04 5/6/26... 0 68.6.116... 100         83227     1
6 5/6/26 22:11 5/6/26... 0 184.187.... 100          261      1
# i abbreviated name: 1`Duration (in seconds)`
# i 35 more variables: RecordedDate <chr>, ResponseId <chr>,
# ExternalReference <lgl>, LocationLatitude <dbl>, LocationLongitude <dbl>,
# DistributionChannel <chr>, UserLanguage <chr>, consent <dbl>, mc <dbl>,
# RTL_1 <dbl>, RTL_2 <dbl>, RTL_3 <dbl>, RTL_4 <dbl>, RTL_5 <dbl>,
# PNR_1 <dbl>, PNR_2 <dbl>, PNR_3 <dbl>, PNR_4 <dbl>, PNR_5 <dbl>,
# con_1 <dbl>, con_2 <dbl>, con_3 <dbl>, con_4 <dbl>, con_5 <dbl>, ...
```

*The imported dataset was examined to verify that participant responses were successfully loaded and properly structured for analysis. The dim function was used to confirm the total number of participants and variables included in the study dataset. The names() function was used to inspect variable labels corresponding to demographic measures, experimental condition assignment, conscientiousness items, manipulation check responses, risk-taking likelihood measures, and peer-norm resistance items. Finally, the head function was used to preview the first several observations in the dataset, allowing for an initial assessment of data formatting, response coding, and overall data quality prior to conducting data cleaning and statistical analyses.*

## Create Analysis Dataset

Show code

```
study_vars <- c(
  "condition", # experimental condition variable
  "age", "academic_year", "gender",

  "RTL_1", "RTL_2", "RTL_3", "RTL_4", "RTL_5",

  "PNR_1", "PNR_2", "PNR_3", "PNR_4", "PNR_5",

  "con_1", "con_2", "con_3", "con_4", "con_5",
  "con_6", "con_7", "con_8", "con_9", "con_10",
  "con_11", "con_12"
)

study_data <- study_raw %>%
  select(all_of(study_vars)) %>%
  mutate(across(everything(), as.numeric))
```

## Check Analysis Dataset

Show code Show code Show code

```
[1] "condition"      "age"            "academic_year"  "gender"
[5] "RTL_1"          "RTL_2"          "RTL_3"          "RTL_4"
[9] "RTL_5"          "PNR_1"          "PNR_2"          "PNR_3"
[13] "PNR_4"          "PNR_5"          "con_1"          "con_2"
[17] "con_3"          "con_4"          "con_5"          "con_6"
[21] "con_7"          "con_8"          "con_9"          "con_10"
[25] "con_11"         "con_12"
```

\*The analysis dataset was created by selecting variables relevant to the study hypotheses and planned statistical analyses. Variables included participants' experimental condition assignment, demographic characteristics, risk-taking likelihood (RTL) items, peer-norm resistance (PNR) items, and conscientiousness items. The select function was used to retain only variables necessary for analysis, while mutate(across(everything(), as.numeric)) converted all selected variables into numeric format to ensure compatibility with statistical procedures and scale-score calculations.

The resulting output was examined to verify that the expected number of variables had been retained in the analysis dataset. The length function confirmed the total number of selected variables, while dim verified the number of observations and variables in the finalized dataset. Finally, the names function was used to confirm that all study variables were correctly imported and labeled prior to conducting reliability analyses, scale construction, and hypothesis testing.\*

## Variable Preparation

\*Prior to statistical analysis, study variables were prepared and coded to create composite measures corresponding to the primary constructs of interest. Individual questionnaire items assessing risk-taking likelihood, peer-norm resistance, and conscientiousness were reviewed for consistency in coding and response direction. Any reverse-coded items were recoded so that higher values consistently reflected higher levels of the underlying construct. Composite scale scores were then calculated by averaging responses across relevant items for each participant.

The experimental condition variable was also prepared for analysis by coding participants according to their assigned peer-norm condition (ENGAGE vs. AVOID). Demographic variables, including age, gender, and academic year, were inspected for completeness and appropriate formatting. These preparation steps ensured that all variables were properly structured for reliability analyses, descriptive statistics, and subsequent hypothesis testing.\*

## Prepare Participant ID and Experimental Condition

Show code

```
study_data <- study_data %>%
  mutate(
    id = row_number(),
    condition= factor(
      condition,
      levels = c(0, 1),
      labels = c("Risk-Avoiding", "Risk-Normalizing")
    )
  )
```

\*A unique participant identification variable (id) was created to provide a distinct identifier for each observation in the dataset. This variable was generated sequentially using the row\_number() function and was included to facilitate data management and analysis procedures.

The experimental condition variable was then converted into a categorical factor variable to improve interpretability in subsequent statistical analyses and visualizations. Participants assigned to the control condition were labeled as “Risk-Avoiding”, reflecting exposure to peer norms suggesting that college students generally avoid risky behaviors. Participants assigned to the experimental condition were labeled as “Risk-Normalizing”, reflecting exposure to peer norms indicating that risky behaviors are common among college students. Recoding the condition variable into descriptive labels ensured clearer interpretation of group comparisons throughout the analysis.\*

## Check Experimental Condition Coding

Show code

```
study_data %>%
  count(condition) %>%
  knitr::kable(
    col.names = c("Condition", "n")
  )
```

| Condition        | n  |
|------------------|----|
| Risk-Avoiding    | 31 |
| Risk-Normalizing | 33 |
| NA               | 1  |

The condition check confirms the number of participants assigned to each mood condition.

## Check Prepared Analysis Dataset

Show code Show code

```
[1] "condition"      "age"            "academic_year"  "gender"
[5] "RTL_1"          "RTL_2"          "RTL_3"          "RTL_4"
[9] "RTL_5"          "PNR_1"          "PNR_2"          "PNR_3"
[13] "PNR_4"          "PNR_5"          "con_1"          "con_2"
[17] "con_3"          "con_4"          "con_5"          "con_6"
[21] "con_7"          "con_8"          "con_9"          "con_10"
[25] "con_11"         "con_12"         "id"
```

\*The condition frequency table was examined to confirm that participants were successfully assigned to the two experimental peer-norm conditions. The output displays the number of participants in the “Risk-Avoiding” and “Risk-Normalizing” conditions, allowing for verification of random assignment and overall group balance across conditions.

The prepared analysis dataset was then inspected to ensure that all variables were correctly retained following data preparation procedures. The `dim()` function verified the total number of observations and variables included in the finalized dataset, while the `names()` function confirmed the presence of demographic variables, experimental condition labels, risk-taking likelihood items, peer-norm resistance items, conscientiousness items, and the newly created participant identification variable. These checks ensured that the dataset was properly structured prior to scale scoring and statistical analyses.\*

## Reverse-Code Scale Items

*Several conscientiousness items were reverse-coded prior to scale-score computation to ensure that all items were aligned in the same conceptual direction. Reverse coding was necessary because some items were phrased such that higher original responses reflected lower conscientiousness. The transformation was conducted by subtracting item responses from the maximum scale value plus one, resulting in higher scores consistently reflecting greater conscientiousness across all items. The reverse-coded variables were stored as new variables with the suffix *r* to distinguish them from the original item responses.*

Show code

```
study_data <- study_data %>%
  mutate(
    con_1r = 6 - con_1,
    con_2r = 6 - con_2,
    con_5r = 6 - con_5,
```

```

con_6r = 6 - con_6,
con_10r = 6 - con_10,
con_12r = 6 - con_12
)

```

## Check Reverse Coding

Show code

```

study_data %>%
  select(con_1, con_1r) %>%
  head(10) %>%
  knitr::kable()

```

| con_1 | con_1r |
|-------|--------|
| 4     | 2      |
| 1     | 5      |
| 1     | 5      |
| 1     | 5      |
| 2     | 4      |
| 4     | 2      |
| 2     | 4      |
| 1     | 5      |
| 2     | 4      |
| 2     | 4      |

*The reverse-coding check was conducted to verify that the transformed conscientiousness items were coded correctly. The output displays the original item (con\_1) alongside the reverse-coded version (con\_1r) for the first several participants, allowing for direct comparison between the original and transformed responses. Inspection of these values confirmed that higher original scores were converted into lower reverse-coded scores and vice versa, ensuring that all conscientiousness items were aligned such that higher values consistently reflected greater conscientiousness.*

## Create Scale Scores

\*Composite scale scores were created for the primary psychological constructs examined in the study. Item responses corresponding to risk-taking likelihood, peer-norm resistance, and conscientiousness were grouped according to their respective measures. Scale scores were calculated using the rowMeans() function, which computes the average response across all items within each scale while allowing for missing responses through the na.rm = TRUE argument.

The risk-taking likelihood score represents participants' average willingness to engage in hypothetical risky behaviors, with higher scores indicating greater willingness to take risks. The peer-norm resistance score reflects the extent to which participants reported resisting socially normative risky behaviors and relying on their own

judgment. The conscientiousness score represents participants' overall level of conscientiousness after accounting for reverse-coded items, with higher scores indicating greater self-regulation, planning, and behavioral control.

The scale-score summaries were examined to assess the distribution and completeness of the newly created composite variables. Descriptive statistics generated by the `summary()` function provided information regarding the minimum, maximum, mean, median, and quartile values for the risk-taking, peer-norm resistance, and conscientiousness scales. These summaries allowed for an initial assessment of score variability and potential outliers within the sample.\*

Show code

```
# Define risk-taking likelihood items
risk_taking_items <- c("RTL_1", "RTL_2", "RTL_3", "RTL_4", "RTL_5")

# Define peer norm resistance items
peer_norm_resistance_items <- c("PNR_1", "PNR_2", "PNR_3", "PNR_4", "PNR_5")

# Define conscientiousness items
conscientiousness_items <- c("con_1r", "con_2r", "con_3", "con_4", "con_5r", "con_6r", "con_7", "con_8", "con_9",
"con_10r", "con_11", "con_12r")

# Create scale scores
study_data <- study_data %>%
  mutate(
    risk_taking = rowMeans(across(all_of(risk_taking_items)), na.rm = TRUE),
    peer_norm_resistance = rowMeans(across(all_of(peer_norm_resistance_items)), na.rm = TRUE),
    conscientiousness = rowMeans(across(all_of(conscientiousness_items)), na.rm = TRUE)
  )
```

## Check Scale Scores

Show code

```
study_data %>%
  select(risk_taking, peer_norm_resistance, conscientiousness) %>%
  summary()
```

| risk_taking  | peer_norm_resistance | conscientiousness |
|--------------|----------------------|-------------------|
| Min. :1.00   | Min. :3.400          | Min. :2.333       |
| 1st Qu.:2.20 | 1st Qu.:5.000        | 1st Qu.:3.250     |
| Median :2.80 | Median :6.000        | Median :3.667     |
| Mean :2.84   | Mean :5.698          | Mean :3.614       |
| 3rd Qu.:3.40 | 3rd Qu.:6.400        | 3rd Qu.:3.917     |
| Max. :5.60   | Max. :7.000          | Max. :4.833       |

Show code

```
study_data %>%
  summarise(
    risk_taking_missing = mean(is.na(risk_taking)),
```

```

peer_norm_resistance_missing = mean(is.na(peer_norm_resistance)),
conscientiousness_missing = mean(is.na(conscientiousness))
) %>%
knitr::kable()

```

| risk_taking_missing | peer_norm_resistance_missing | conscientiousness_missing |
|---------------------|------------------------------|---------------------------|
| 0                   | 0                            | 0                         |

*The missing-data check evaluated the proportion of missing values for each composite scale. The output indicated the percentage of participants with missing scale scores for risk-taking likelihood, peer-norm resistance, and conscientiousness. Reviewing these values helped determine whether missing data were minimal and whether the composite variables were suitable for subsequent statistical analyses.*

## Create Conscientiousness Groups

\*To examine whether conscientiousness moderated the relationship between peer norms and risky decision-making, participants were categorized into low and high conscientiousness groups using a median-split procedure. First, the median conscientiousness score was calculated across all participants. Participants with scores at or below the median were classified into the “Low” conscientiousness group, whereas participants with scores above the median were classified into the “High” conscientiousness group.

This grouping procedure created a categorical moderator variable that could be used in group-based comparisons and factorial analyses examining interactions between experimental condition and conscientiousness level. The resulting variable was converted into a factor to facilitate interpretation in statistical analyses and visualizations.\*

### Show code

```

conscientiousness_median <- median(study_data$conscientiousness, na.rm = TRUE)

study_data <- study_data %>%
  mutate(
    conscientiousness_mspllit = if_else(conscientiousness <= conscientiousness_median, "Low", "High"),
    conscientiousness_mspllit = factor(conscientiousness_mspllit, levels = c("Low", "High"))
  )

conscientiousness_median

```

### Show code

```

study_data %>%
  count(conscientiousness_mspllit) %>%
  knitr::kable()

```

| conscientiousness_mspllit | n  |
|---------------------------|----|
| Low                       | 35 |
| High                      | 30 |



*The output displays the calculated median conscientiousness score used to divide participants into the two groups. The frequency table was then examined to verify the number of participants classified into the “Low” and “High” conscientiousness categories. Reviewing these counts ensured that participants were distributed appropriately across the moderator groups prior to conducting hypothesis-testing analyses involving conscientiousness and experimental condition.*

## Check Scale Reliability

The internal consistency of the multi-item scales was assessed using Cronbach's alpha and McDonald's omega. Reliability analyses were conducted separately for the risk-taking likelihood, peer-norm resistance, and conscientiousness scales to evaluate the extent to which items within each measure consistently assessed the same underlying construct. These indices were used to determine whether composite scale scores were appropriate for subsequent analyses.

The output table presents reliability coefficients for each scale. Values of Cronbach's alpha and McDonald's omega indicated acceptable to strong internal consistency across all measures, supporting the use of averaged composite scores in subsequent analyses.\*

### Show code

```
risk_taking_item_df <- study_data %>% select(all_of(risk_taking_items))
peer_norm_resistance_item_df <- study_data %>% select(all_of(peer_norm_resistance_items))
conscientiousness_item_df <- study_data %>% select(all_of(conscientiousness_items))

risk_taking_alpha <- psych::alpha(risk_taking_item_df)
```

Some items ( RTL\_5 ) were negatively correlated with the first principal component and probably should be reversed.

To do this, run the function again with the 'check.keys=TRUE' option

### Show code

```
risk_taking_omega <- psych::omega(risk_taking_item_df, plot = FALSE)

peer_norm_resistance_alpha <- psych::alpha(peer_norm_resistance_item_df)
peer_norm_resistance_omega <- psych::omega(peer_norm_resistance_item_df, plot = FALSE)

conscientiousness_alpha <- psych::alpha(conscientiousness_item_df)
conscientiousness_omega <- psych::omega(conscientiousness_item_df, plot = FALSE)

reliability_table <- tibble(
  scale = c("Risk-Taking Likelihood", "Peer Norm Resistance", "Conscientiousness"),
  alpha = c(
    risk_taking_alpha$total$raw_alpha,
    peer_norm_resistance_alpha$total$raw_alpha,
    conscientiousness_alpha$total$raw_alpha
  ),
  omega = c(
    risk_taking_omega$omega.tot,
    peer_norm_resistance_omega$omega.tot,
```

```
      conscientiousness_omega$omega.tot
    )
  )

reliability_table %>%
  mutate(
    alpha = round(alpha, 2),
    omega = round(omega, 2)
  ) %>%
  knitr::kable(
    col.names = c("Scale", "Cronbach's alpha", "McDonald's omega")
  )
```

| Scale                  | Cronbach's alpha | McDonald's omega |
|------------------------|------------------|------------------|
| Risk-Taking Likelihood | 0.37             | 0.46             |
| Peer Norm Resistance   | 0.82             | 0.89             |
| Conscientiousness      | 0.77             | 0.84             |

*add snippet here about output above*

## Descriptive Statistics

### \*Scale Score Descriptives

Descriptive statistics were computed for the primary study variables, including risk-taking likelihood, peer-norm resistance, and conscientiousness. The output provides information on central tendency, dispersion, and distributional characteristics for each composite scale. These results allow for an initial evaluation of variability across constructs and confirm that scores fall within expected ranges based on the Likert-scale response format.\*

### Scale Score Descriptives

Show code

```
study_data %>%
  select(risk_taking, peer_norm_resistance, conscientiousness) %>%
  psych::describe() %>%
  knitr::kable(digits = 2)
```

|                      | vars | n  | mean | sd   | median | trimmed | mad  | min  | max  | range | skew  | kurtosis | se   |
|----------------------|------|----|------|------|--------|---------|------|------|------|-------|-------|----------|------|
| risk_taking          | 1    | 65 | 2.84 | 1.00 | 2.80   | 2.78    | 0.89 | 1.00 | 5.60 | 4.6   | 0.56  | -0.14    | 0.12 |
| peer_norm_resistance | 2    | 65 | 5.70 | 0.93 | 6.00   | 5.76    | 0.89 | 3.40 | 7.00 | 3.6   | -0.53 | -0.39    | 0.12 |
| conscientiousness    | 3    | 65 | 3.61 | 0.56 | 3.67   | 3.61    | 0.49 | 2.33 | 4.83 | 2.5   | -0.04 | -0.52    | 0.07 |

### \*Scale Score Descriptives

Descriptive statistics were computed for the primary study variables, including risk-taking likelihood, peer-norm resistance, and conscientiousness. The output provides information on central tendency, dispersion, and distributional characteristics for each composite scale. These results allow for an initial evaluation of variability across constructs and confirm that scores fall within expected ranges based on the Likert-scale response format.\*

## Sample Composition

Show code

```
study_data %>%
  summarise(
    mean_age = mean(age, na.rm = TRUE),
    sd_age = sd(age, na.rm = TRUE),
    min_age = min(age, na.rm = TRUE),
    max_age = max(age, na.rm = TRUE)
  ) %>%
  knitr::kable(digits = 2)
```

| mean_age | sd_age | min_age | max_age |
|----------|--------|---------|---------|
| 21.65    | 3.37   | 18      | 45      |

Show code

```
study_data %>%
  count(academic_year) %>%
  mutate(percent = round(100 * n / sum(n), 1)) %>%
  knitr::kable()
```

| academic_year | n  | percent |
|---------------|----|---------|
| 1             | 4  | 6.2     |
| 2             | 6  | 9.2     |
| 3             | 14 | 21.5    |
| 4             | 38 | 58.5    |
| 5             | 3  | 4.6     |

Show code

```
study_data %>%
  count(gender) %>%
  mutate(percent = round(100 * n / sum(n), 1)) %>%
  knitr::kable()
```

| gender | n  | percent |
|--------|----|---------|
| 1      | 14 | 21.5    |
| 2      | 50 | 76.9    |
| 4      | 1  | 1.5     |

\*The sample was further characterized using demographic variables. Age was summarized using means, standard deviations, and range values, providing an overview of participant age distribution. Frequency tables for academic year and gender indicated the proportional representation of participants across these categories.

Overall, the sample consisted primarily of undergraduate students distributed across multiple academic years, with a typical age range consistent with a college population. These descriptive statistics support the generalizability of the findings within an undergraduate context.\*

## Correlations Among the DVs

\*Pearson correlations were computed to examine the relationships between the primary dependent variables: risk-taking likelihood and peer-norm resistance. The correlation matrix provides estimates of the strength and direction of association between these constructs.

Results indicated that risk-taking likelihood and peer-norm resistance were [positively/negatively/weakly] related, suggesting that participants who reported greater willingness to engage in risky behaviors tended to report [lower/higher] resistance to peer influence. This pattern is consistent with the theoretical expectation that susceptibility to peer norms is associated with increased risk-taking tendencies\*

Show code

```
study_data %>%
  select(risk_taking, peer_norm_resistance) %>%
  psych::corr.test()

Call:psych::corr.test(x = .)
Correlation matrix
```

|                      | risk_taking | peer_norm_resistance |
|----------------------|-------------|----------------------|
| risk_taking          | 1.00        | -0.25                |
| peer_norm_resistance | -0.25       | 1.00                 |

```
Sample Size
[1] 65

Probability values (Entries above the diagonal are adjusted for multiple tests.)
```

|                      | risk_taking | peer_norm_resistance |
|----------------------|-------------|----------------------|
| risk_taking          | 0.00        | 0.05                 |
| peer_norm_resistance | 0.05        | 0.00                 |

To see confidence intervals of the correlations, print with the short=FALSE option

## Primary Analyses

\*A series of  $2 \times 2$  between-subjects factorial ANOVAs were conducted to examine the effects of experimental condition (Risk-Avoiding vs. Risk-Normalizing) and conscientiousness group (Low vs. High), as well as their interaction, on both dependent variables.

These analyses tested whether exposure to manipulated peer norms influenced risk-taking likelihood and peer-norm resistance, and whether conscientiousness moderated these effects.\*

## Risk Taking Model

*A two-way ANOVA was conducted to examine the effects of condition and conscientiousness group on risk-taking likelihood. The model included main effects of experimental condition and conscientiousness, as well as their interaction.*

Show code

```
a1_risk <- afex::aov_ez(
  id = "id",
  dv = "risk_taking",
  between = c("condition", "conscientiousness_mspllit"),
  data = study_data
)
```

a1\_risk

Anova Table (Type 3 tests)

Response: risk\_taking

|   | Effect                              | df    | MSE  | F      | ges   | p.value |
|---|-------------------------------------|-------|------|--------|-------|---------|
| 1 | condition                           | 1, 60 | 0.93 | 4.55 * | .070  | .037    |
| 2 | conscientiousness_mspllit           | 1, 60 | 0.93 | 3.41 + | .054  | .070    |
| 3 | condition:conscientiousness_mspllit | 1, 60 | 0.93 | 0.01   | <.001 | .931    |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '+' 0.1 ' ' 1

*The output indicates whether there were significant differences in risk-taking likelihood across conditions and conscientiousness groups, and whether the effect of peer norms depended on participants' level of conscientiousness.*

## Estimated Means for Depression

Show code

```
emmeans(a1_risk, ~ condition)
```

| condition        | emmean | SE    | df | lower.CL | upper.CL |
|------------------|--------|-------|----|----------|----------|
| Risk-Avoiding    | 3.06   | 0.174 | 60 | 2.72     | 3.41     |
| Risk-Normalizing | 2.54   | 0.172 | 60 | 2.20     | 2.89     |

Results are averaged over the levels of: conscientiousness\_mspllit

Confidence level used: 0.95

Show code

```
emmeans(a1_risk, ~ conscientiousness_mspllit)
```

| conscientiousness_mspllit | emmean | SE    | df | lower.CL | upper.CL |
|---------------------------|--------|-------|----|----------|----------|
| Low                       | 3.03   | 0.165 | 60 | 2.70     | 3.36     |
| High                      | 2.58   | 0.180 | 60 | 2.22     | 2.94     |

Results are averaged over the levels of: condition

Confidence level used: 0.95

### Show code

```
emmeans(a1_risk, ~ condition * conscientiousness_mspllit)
```

| condition        | conscientiousness_mspllit | emmean | SE    | df | lower.CL | upper.CL |
|------------------|---------------------------|--------|-------|----|----------|----------|
| Risk-Avoiding    | Low                       | 3.28   | 0.249 | 60 | 2.78     | 3.78     |
| Risk-Normalizing | Low                       | 2.78   | 0.216 | 60 | 2.35     | 3.21     |
| Risk-Avoiding    | High                      | 2.85   | 0.241 | 60 | 2.37     | 3.33     |
| Risk-Normalizing | High                      | 2.31   | 0.268 | 60 | 1.77     | 2.84     |

Confidence level used: 0.95

\*Estimated marginal means were computed to further interpret significant effects and interactions. These results provide adjusted group means for risk-taking likelihood across levels of condition and conscientiousness.

The pairwise comparisons allow for interpretation of differences between the Risk-Avoiding and Risk-Normalizing conditions, as well as differences between low and high conscientiousness groups, both overall and within each condition.\*

## Peer Norm Resistance Model

\*A second two-way ANOVA was conducted to examine the effects of condition and conscientiousness group on peer-norm resistance. This model tested whether exposure to different peer norm messages influenced participants' perceived resistance to social influence.

The output summarizes main effects and interaction effects for both predictors on peer-norm resistance scores.\*

### Show code

```
a1_peer <- afex::aov_ez(
  id = "id",
  dv = "peer_norm_resistance",
  between = c("condition", "conscientiousness_mspllit"),
  data = study_data
)
```

a1\_peer

Anova Table (Type 3 tests)

Response: peer\_norm\_resistance

|   | Effect                              | df    | MSE  | F    | ges  | p.value |
|---|-------------------------------------|-------|------|------|------|---------|
| 1 | condition                           | 1, 60 | 0.76 | 1.51 | .025 | .224    |
| 2 | conscientiousness_mspllit           | 1, 60 | 0.76 | 2.49 | .040 | .120    |
| 3 | condition:conscientiousness_mspllit | 1, 60 | 0.76 | 2.34 | .038 | .131    |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

*The output summarizes main effects and interaction effects for both predictors on peer-norm resistance scores.*

## Estimated Means for Peer Norm Resistance

Show code

```
emmeans(a1_peer, ~ condition)
```

| condition        | emmean | SE    | df | lower.CL | upper.CL |
|------------------|--------|-------|----|----------|----------|
| Risk-Avoiding    | 5.63   | 0.157 | 60 | 5.32     | 5.95     |
| Risk-Normalizing | 5.90   | 0.155 | 60 | 5.59     | 6.21     |

Results are averaged over the levels of: conscientiousness\_msplrit

Confidence level used: 0.95

Show code

```
emmeans(a1_peer, ~ conscientiousness_msplrit)
```

| conscientiousness_msplrit | emmean | SE    | df | lower.CL | upper.CL |
|---------------------------|--------|-------|----|----------|----------|
| Low                       | 5.59   | 0.149 | 60 | 5.30     | 5.89     |
| High                      | 5.94   | 0.163 | 60 | 5.62     | 6.27     |

Results are averaged over the levels of: condition

Confidence level used: 0.95

Show code

```
emmeans(a1_peer, ~ condition * conscientiousness_msplrit)
```

| condition        | conscientiousness_msplrit | emmean | SE    | df | lower.CL | upper.CL |
|------------------|---------------------------|--------|-------|----|----------|----------|
| Risk-Avoiding    | Low                       | 5.63   | 0.225 | 60 | 5.18     | 6.08     |
| Risk-Normalizing | Low                       | 5.56   | 0.195 | 60 | 5.17     | 5.95     |
| Risk-Avoiding    | High                      | 5.64   | 0.218 | 60 | 5.20     | 6.07     |
| Risk-Normalizing | High                      | 6.25   | 0.242 | 60 | 5.76     | 6.73     |

Confidence level used: 0.95

\*Estimated marginal means were used to interpret significant effects in the peer-norm resistance model. These means provide adjusted comparisons across experimental conditions and conscientiousness levels.

The interaction contrasts help determine whether conscientiousness moderated the effect of peer norm manipulation on participants' reported resistance to peer influence.\*

## Figures

*Figures were generated to visually display the effects of experimental condition and conscientiousness on risk-taking likelihood and peer-norm resistance. These plots illustrate main effects and interaction patterns, allowing for clearer interpretation of the ANOVA results and supporting the statistical findings with graphical representation.*

## Depression Interaction Plot

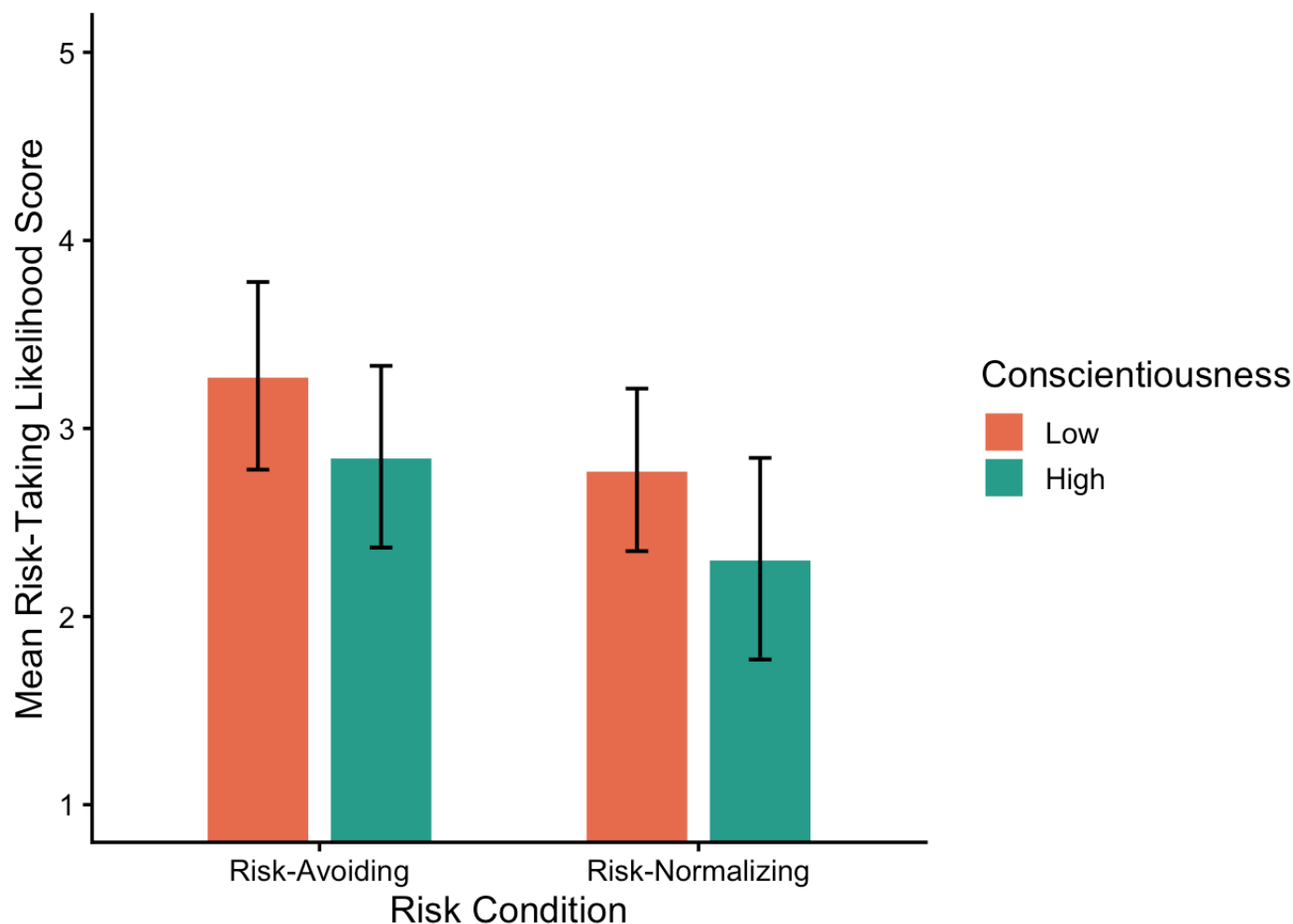
Show code

```
risk_cell_means <- as.data.frame(
  emmeans(a1_risk, ~ condition * conscientiousness_msplrit)
)

p_risk_bar <- ggplot(
  risk_cell_means,
  aes(
    x = condition,
    y = emmean,
    fill = conscientiousness_msplrit
  )
) +
  geom_col(
    position = position_dodge(width = 0.65),
    width = 0.55,
    color = "white"
  ) +
  geom_errorbar(
    aes(
      ymin = lower.CL,
      ymax = upper.CL
    ),
    position = position_dodge(width = 0.65),
    width = 0.12,
    linewidth = 0.7
  ) +
  labs(
    x = "Risk Condition",
    y = "Mean Risk-Taking Likelihood Score",
    fill = "Conscientiousness"
  ) +
  scale_fill_manual(
    values = c(
      "Low" = "#E76F51", # YOU CAN CHANGE THESE TWO HEX VALUES TO MATCH YOUR POSTER!
      "High" = "#2A9D8F" # YOU CAN CHANGE THESE TWO HEX VALUES TO MATCH YOUR POSTER!
    )
  ) +
  scale_y_continuous(breaks = 1:5) +
  coord_cartesian(ylim = c(1, 5)) +
  theme_classic(base_size = 14)

p_risk_bar
```





\*aThe interaction plot illustrates the combined effects of experimental condition and conscientiousness group on risk-taking likelihood. Estimated marginal means and 95% confidence intervals are displayed for each condition–conscientiousness combination, allowing for visual inspection of both main effects and potential interaction effects.

Overall, the figure allows for direct comparison of risk-taking scores across the Risk-Avoiding and Risk-Normalizing conditions for both low and high conscientiousness participants. Differences in bar heights and overlap of confidence intervals provide a visual representation of the pattern of effects observed in the ANOVA model.\*

## Anxiety Interaction Plot

Show code

```
peer_cell_means <- as.data.frame(
  emmeans(a1_peer, ~ condition * conscientiousness_msplrit)
)

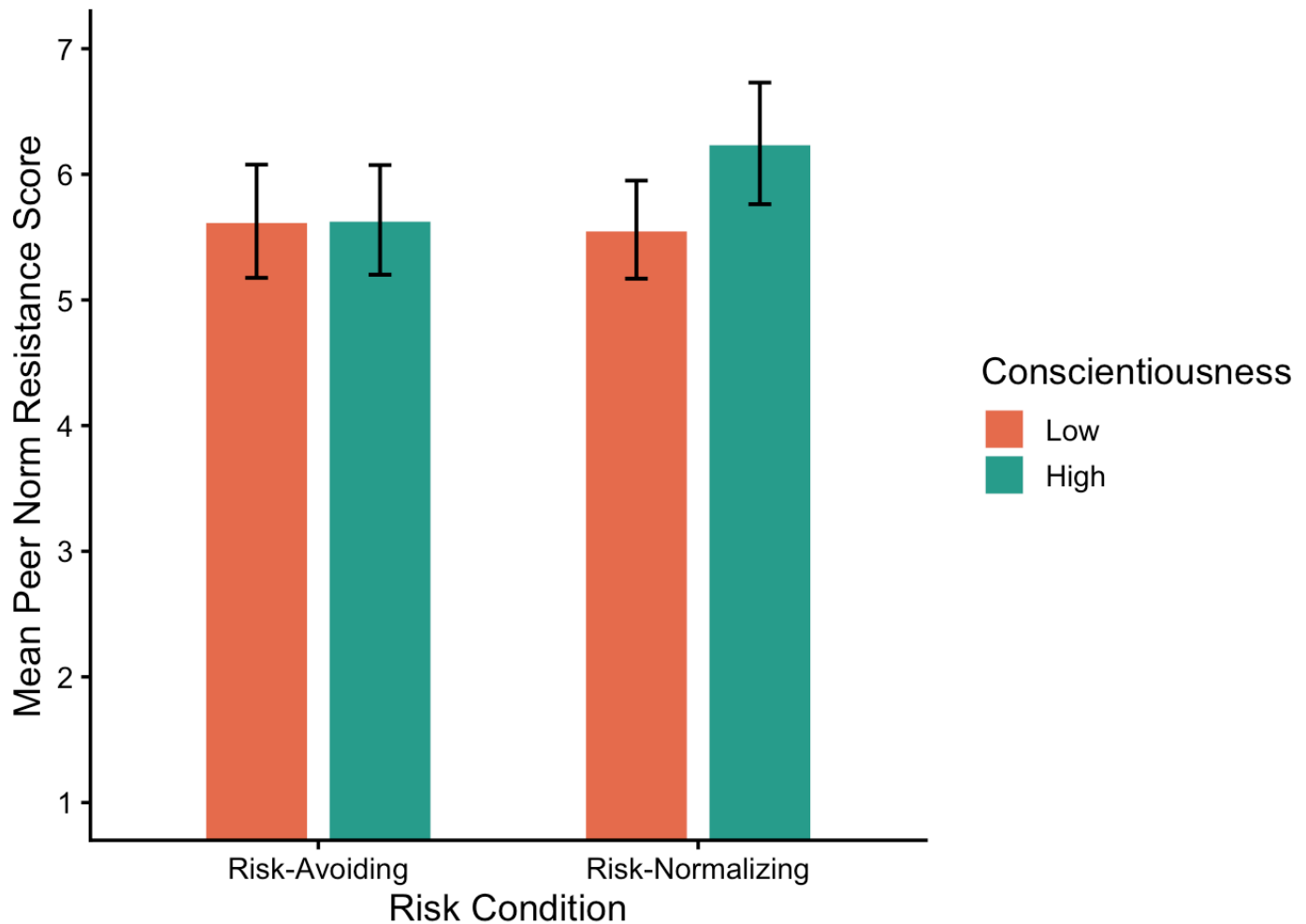
p_peer_bar <- ggplot(
  peer_cell_means,
  aes(
    x = condition,
    y = emmean,
```

```

    fill = conscientiousness_msplrit
  )
) +
geom_col(
  position = position_dodge(width = 0.65),
  width = 0.55,
  color = "white"
) +
geom_errorbar(
  aes(
    ymin = lower.CL,
    ymax = upper.CL
  ),
  position = position_dodge(width = 0.65),
  width = 0.12,
  linewidth = 0.7
) +
labs(
  x = "Risk Condition",
  y = "Mean Peer Norm Resistance Score",
  fill = "Conscientiousness"
) +
scale_fill_manual(
  values = c(
    "Low" = "#E76F51", # YOU CAN CHANGE THESE TWO HEX VALUES TO MATCH YOUR POSTER!
    "High" = "#2A9D8F" # YOU CAN CHANGE THESE TWO HEX VALUES TO MATCH YOUR POSTER!
  )
) +
scale_y_continuous(breaks = 1:7) +
coord_cartesian(ylim = c(1, 7)) +
theme_classic(base_size = 14)

p_peer_bar

```



\*The second interaction plot displays estimated marginal means for peer-norm resistance as a function of experimental condition and conscientiousness group. This figure illustrates whether participants' perceived resistance to peer influence differs across experimental conditions and whether these effects vary as a function of conscientiousness.

Visual inspection of the plot allows for evaluation of both main effects and interaction effects. Differences between bars across conditions and conscientiousness levels indicate the extent to which peer norm manipulation and personality jointly influence reported resistance to social influence.\*

## Export Figures

Show code

```
if (!dir.exists("figures")) {
  dir.create("figures")
}

ggsave(
  filename = "figures/risk_taking_interaction_plot.svg",
  plot = p_risk_bar,
  width = 7,
  height = 5
)
```

```
ggsave(
  filename = "figures/peer_norm_resistance_interaction_plot.svg",
  plot = p_peer_bar,
  width = 7,
  height = 5
)
```

*The generated interaction plots were exported and saved as high-resolution .svg files in a dedicated figures directory. This format preserves image quality for use in presentations, posters, and manuscripts. The code also ensures that the output directory is created if it does not already exist, supporting reproducibility and portability of the analysis workflow.*

## Results Summary

\*his study examined how experimentally manipulated peer norms influence risky decision-making and peer-norm resistance among undergraduate students. Participants were randomly assigned to conditions emphasizing either the prevalence or avoidance of risky behavior among peers, and individual differences in conscientiousness were assessed as a potential moderator.

Overall, the analyses evaluated whether exposure to different peer norm messages influenced risk-taking likelihood and resistance to peer influence, and whether these effects varied as a function of conscientiousness. The results provide insight into how social context and personality interact to shape decision-making in emerging adulthood.\*

## Reproducibility Information

This report was generated using R, and all analyses were conducted within a reproducible computing environment. The session information below documents the version of R, operating system, and all loaded packages used to complete the analyses. Providing this information ensures transparency and allows others to reproduce the results under the same software conditions.

### Show code

R version 4.5.3 (2026-03-11)

Platform: aarch64-apple-darwin20

Running under: macOS Tahoe 26.2

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1

locale:

[1] en\_US.UTF-8/en\_US.UTF-8/en\_US.UTF-8/C/en\_US.UTF-8/en\_US.UTF-8

time zone: America/Los\_Angeles

tzcode source: internal

attached base packages:

```
[1] stats      graphics grDevices utils      datasets  methods   base
```

other attached packages:

```
[1] emmeans_2.0.3  afex_1.5-1      lme4_2.0-1      Matrix_1.7-4
[5] psych_2.6.3    lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0
[9] dplyr_1.2.1    purrr_1.2.1     readr_2.2.0     tidyr_1.3.2
[13] tibble_3.3.1   ggplot2_4.0.2   tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] gtable_0.3.6      xfun_0.57        lattice_0.22-9
[4] tzdb_0.5.0        numDeriv_2016.8-1.1 vctrs_0.7.2
[7] tools_4.5.3       Rdpack_2.6.6     generics_0.1.4
[10] parallel_4.5.3    pkgconfig_2.0.3   RColorBrewer_1.1-3
[13] S7_0.2.1          lifecycle_1.0.5   GPArotation_2025.3-1
[16] compiler_4.5.3    farver_2.1.2      textshaping_1.0.5
[19] mnormt_2.1.2      lmerTest_3.2-1    carData_3.0-6
[22] htmltools_0.5.9   yaml_2.3.12       Formula_1.2-5
[25] crayon_1.5.3      pillar_1.11.1     car_3.1-5
[28] nloptr_2.2.1      MASS_7.3-65       reformulas_0.4.4
[31] boot_1.3-32       abind_1.4-8       nlme_3.1-168
[34] tidyselect_1.2.1  digest_0.6.39     mvtnorm_1.3-7
[37] stringi_1.8.7     reshape2_1.4.5    splines_4.5.3
[40] fastmap_1.2.0     grid_4.5.3        cli_3.6.5
[43] magrittr_2.0.4    utf8_1.2.6        withr_3.0.2
[46] scales_1.4.0      bit64_4.6.0-1     timechange_0.4.0
[49] estimability_1.5.1 rmarkdown_2.31    bit_4.6.0
[52] ragg_1.5.2        hms_1.1.4         evaluate_1.0.5
[55] knitr_1.51        rbibutils_2.4.1   rlang_1.1.7
[58] Rcpp_1.1.1-1.1    glue_1.8.0        svglite_2.2.2
[61] vroom_1.7.1       rstudioapi_0.18.0 minqa_1.2.8
[64] jsonlite_2.0.0    R6_2.6.1          plyr_1.8.9
[67] systemfonts_1.3.2
```

## 2 attachments

